

SECTION B

6 (short)

(a) Several 64k bit memory devices, each with four data lines, are to be connected to a microprocessor with an 8-bit data bus.

(i) How many and what kinds of connections will there be on each memory device? [3]

(ii) How many devices are needed to make 64k bytes of memory? [1]

(b) (i) What is a shift register? [3]

(ii) The outputs from the final two stages of a 4-stage shift register are fed back to the input to the first stage via an exclusive-OR gate. How many steps are there in the longest sequence of the shift register states? [3]